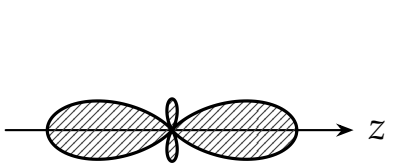
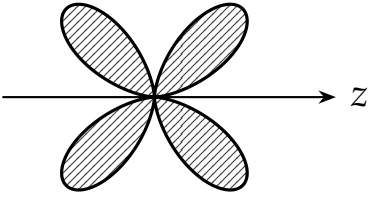
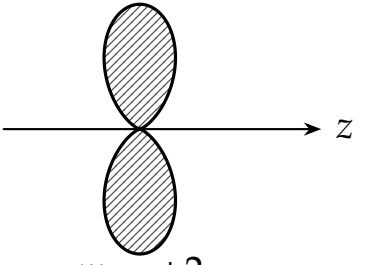


$m = 0$



$m = \pm 1$



$m = \pm 2$